SHAIK AKRAM

Java Backend Developer

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Professional Summary

Java backend developer with an MCA and 9.0 CGPA, experienced in building REST APIs and distributed backend services using Java 17, Spring Boot, Kafka, Redis, and MySQL. Developed a Kafka-driven workflow orchestration platform with distributed locking, retry recovery, and dead-letter queues. Improved book-listing throughput from approximately 30 to 160 requests per second through Redis caching. Experienced in JWT authentication, role-based access control, payment integration, Docker, and automated testing.

Technical Skills

Languages: Java 17, SQL

Backend Development: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST APIs, JWT, Role-Based Access Control

Microservices and Messaging: Spring Cloud, Eureka, Spring Cloud Gateway, Resilience4j, Apache Kafka, Consumer Groups, Dead Letter Queues, Retry and Recovery

Databases and Caching: MySQL, Redis

Testing and Performance: JUnit 5, Mockito, Testcontainers, MockMvc, k6

Tools and DevOps: Docker, Docker Compose, Git, Maven, Swagger/OpenAPI

Problem Solving: Data Structures and Algorithms, 180+ LeetCode problems solved (82 Medium, 11 Hard)

Projects

FlowForge: Distributed Workflow Orchestration Platform GitHub | Architecture Diagram

Java 17, Spring Boot, Apache Kafka, Redis, MySQL, Eureka, Spring Cloud Gateway, Resilience4j, Docker, Testcontainers

- Designed a Kafka-driven workflow orchestration engine using Spring Boot microservices, with DFS-based cycle detection to reject circular workflow definitions.
- Implemented Redis distributed locking with SET NX and Lua-based token-verified unlocking to prevent concurrent workers from executing the same task simultaneously.
- Developed exponential-backoff retries, dead-letter queue routing, and scheduled recovery for tasks stuck in the DISPATCHED state.
- Tested failure recovery in Docker by terminating workers during execution and interrupting Kafka, verifying recovery behavior and preventing duplicate task completion.
- Implemented horizontal worker scaling using Kafka consumer groups and added 30 unit tests and 2 Testcontainers integration tests.

Library Management System: REST API GitHub | Architecture Diagram

Java 17, Spring Boot, Spring Security, JWT, MySQL, Redis, Docker, JUnit 5, Mockito, Swagger/OpenAPI

- Built 45+ REST endpoints for books, authors, members, borrowing, and authentication, with ADMIN, LIBRARIAN, and USER roles enforced through Spring Security and @PreAuthorize.
- Implemented JWT access and refresh tokens, refresh-token rotation, logout blacklisting, and a secure token refresh endpoint.
- Benchmarked performance using k6 with 50 virtual users for 60 seconds. Redis caching increased throughput from approximately 30 to 160 requests per second, reduced p95 latency from 2.9 seconds to under 0.4 seconds, and decreased MySQL row reads by approximately 96%.
- Added scheduled reports, overdue reminders, and token cleanup. Containerized the application with Docker Compose and wrote 30+ tests using JUnit 5, Mockito, MockMvc, and DataJpaTest.

E-Commerce Backend: REST API GitHub | Architecture Diagram

Java 17, Spring Boot 3.5.4, Spring Security, JWT, MySQL, Redis, Razorpay, JUnit 5, Mockito, Swagger/OpenAPI

- Developed 50+ REST endpoints covering users, products, carts, orders, payments, reviews, wishlists, and analytics using Java 17 and Spring Boot.
- Integrated Razorpay payment processing with order creation, payment verification, transaction tracking, and webhook handling.
- Implemented JWT authentication with access and refresh token rotation, logout blacklisting, and BCrypt password hashing.
- Added Redis caching for products, categories, carts, and user profiles, with cache eviction on updates. Implemented sales analytics, top-product reports, low-stock tracking, and asynchronous email notifications.

Internship Experience

Python Developer Intern *2024 | 3 Months*
Synchro Serve Global Solution

- Automated daily CSV-to-MySQL ingestion, reducing execution time from 2 hours to 4 minutes.
- Developed and tested 5 Python utilities for data validation, duplicate detection, and MySQL audit logging.
- Identified SQL injection risks in two scripts and replaced string-concatenated queries with parameterized queries.

Data Analytics Intern *2025 | 3 Months*
Sri Vensy Technologies

- Cleaned and transformed 5 datasets using Python and Excel and developed 4 Tableau dashboards for sales and operational KPIs.
- Identified redundant data-processing steps and contributed recommendations adopted by the analytics team.

Education

Master of Computer Applications (MCA), VEMU Institute of Technology (Autonomous) 2024–2026 | CGPA: 9.0/10.0
B.Sc. Mathematics, Physics and Computer Science, Margadarsi Degree College 2021–2024 | CGPA: 8.1/10.0
Intermediate (MPC), Priyadarshini College 2019–2021 | 509 Marks
Secondary School Certificate (SSC), Government High School 2019 | CGPA: 9.8/10.0

Certifications

- Data Analytics Essentials, Cisco Networking Academy
- SQL and Relational Databases, IBM Cognitive Class
- Data Analytics, Sri Vensy Technologies
- Python Programming, Synchro Serve Global Solution
- C Language, GeeksforGeeks